S-RQ1. Supplementary Material for RQ1 (Generation and Execution Efficiency)

This appendix collects the supplementary tables referenced in the RQ1 results section (Section **??**) of the main manuscript. Tables S1 and S2 expand on the mechanism paragraph in Section **??**: the per-component pooled Spearman correlation between edge count and each phase of T_{gen} for *Model Once, Generate Any*, and the median share of the L-sequence transformation in T_{gen}. Table S3 gives the per-SPL Spearman correlation referenced inline for Bank Account, Student Attendance System, and Elevator. Tables S4 and S5 report per-SPL median peak heap memory and median test execution time for every (approach, L) combination; selected values are cited inline in Section **??**. Table S6 gives the run-to-run stability statistics that back the IQR% summary in Section **??**. Table S7 reports the JVM warm-up evidence cited at the end of Section **??** and again in Section **??**: per-cell mean/median and max/median ratios across the eleven repetitions. Table S8 is the full pairwise Wilcoxon table (with Vargha–Delaney A_{12} effect sizes and Benjamini–Hochberg corrected p-values) for every metric and (SPL, L) combination; the manuscript reports only the headline findings. Table S9 gives the per-SPL edge-count distribution that contextualises the per-SPL correlations in Table S3 and the within-SPL discussion in Section **??**.

Spearman ρ values are reported to three decimal places. Percentage values use ROUND_HALF_UP to one decimal place after a six-decimal float-noise scrub. Approach abbreviations: MO = *Model Once, Generate Any*, Struct = structural baseline, Stoch = stochastic baseline.

Table 1
S1: Pooled Spearman rank correlation ρ between per-product edge count and each component of T_{gen} for *Model Once, Generate Any*, computed across all 4,455 products. The increase in component-level ρ with L explains the increase in the overall ρ reported for *Model Once, Generate Any* in Section **??**.

Component	$L = 2$	$L = 3$	$L = 4$
Model loading	0.206	0.191	0.175
L-sequence transformation	0.186	0.377	0.507
Euler traversal (algorithm)	0.386	0.548	0.613
Test recording	0.284	0.451	0.550
Total T_{gen}	0.355	0.526	0.602

Table 2
S2: Median and IQR of the share of the L-sequence transformation in T_{gen} for *Model Once, Generate Any*, computed across all 4,455 products.

Coverage level	**Median share (%)**	**IQR width (%)**
$L = 2$	3.0	3.0
$L = 3$	11.3	11.0
$L = 4$	16.3	14.2

Table 3
S3: Per-SPL Spearman rank correlation ρ between per-product edge count and per-product median T_{gen}, for every (approach, L) combination. The pooled values reported in Section **??** aggregate these per-SPL correlations and are dominated by between-SPL variation in model size.

SPL	*Model Once, Generate Any*			Structural baseline			Stochastic baseline
	$L = 2$	$L = 3$	$L = 4$	$L = 2$	$L = 3$	$L = 4$	—
SVM	0.063	-0.252	0.022	-0.313	-0.360	-0.317	0.374
eM	0.510	0.369	0.467	0.364	0.478	0.333	0.395
El	-0.005	0.323	0.806	-0.141	-0.154	-0.083	0.369
BAv2	-0.021	-0.022	-0.016	0.018	0.018	0.045	0.067
SAS	0.348	0.529	0.619	0.134	0.119	0.076	0.487
Svia	0.289	0.216	0.269	0.101	0.114	0.105	-0.086
Te	0.344	0.430	0.457	0.007	0.040	0.044	0.516
HS	0.183	0.283	0.310	0.055	0.011	0.057	0.160

Table 4
S4: Per-SPL median peak heap memory (MB) during test generation, per (approach, L) combination. Selected values are cited inline in Section **??**.

SPL	*Model Once, Generate Any*			Structural baseline			Stochastic baseline
	$L = 2$	$L = 3$	$L = 4$	$L = 2$	$L = 3$	$L = 4$	—
SVM	9.0	9.0	9.0	8.0	8.0	8.0	6.0
eM	10.0	11.0	15.0	8.0	8.0	9.0	7.0
El	11.0	16.0	31.9	9.0	11.0	12.0	7.0
BAv2	11.0	11.0	11.0	10.0	10.0	10.0	12.1
SAS	15.0	18.0	24.0	15.0	16.0	17.0	19.5
Svia	127.7	260.2	454.1	40.7	43.7	50.7	311.6
Te	13.0	19.7	38.3	13.3	18.3	28.4	33.3
HS	15.0	17.0	20.9	17.0	17.0	18.0	135.5

Table 5
S5: Per-SPL median test execution time T_{exec} (ms), per (approach, L) combination. Across the board, T_{exec} is small relative to T_{gen} (Section **??**).

SPL	*Model Once, Generate Any*			Structural baseline			Stochastic baseline
	$L = 2$	$L = 3$	$L = 4$	$L = 2$	$L = 3$	$L = 4$	—
SVM	9.1	9.2	9.6	11.1	10.7	10.3	11.0
eM	19.3	22.9	23.3	18.2	19.8	24.4	24.1
El	34.0	36.8	39.2	44.8	62.2	73.6	53.4
BAv2	128.8	124.9	101.4	242.1	223.0	192.1	131.9
SAS	648.2	401.7	449.7	759.6	907.6	1047.9	916.5
Svia	1064.0	466.1	376.5	193.0	361.6	598.3	9584.0
Te	256.3	125.1	125.4	434.5	473.4	528.7	341.8
HS	310.8	207.8	201.2	344.7	435.3	500.5	487.4

Table 6
S6: Run-to-run stability of T_{gen} across the eleven repetitions of each (SPL, approach, L) cell. Median, min, max, coefficient of variation $\mathrm{CV}\% = 100 \cdot s/\bar{x}$, and the relative interquartile range $\mathrm{IQR}\% = 100 \cdot (Q_3 - Q_1)/\text{median}$ are reported for each cell. Section **??** summarises these distributions.

Approach	SPL	L	N	Median (ms)	Min (ms)	Max (ms)	CV (%)	IQR % of median
Model Once, Generate Any	SVM	2	11	1389.6	1165.8	1913.3	15.8	15.4
	eM	2	11	2773.9	2459.9	3092.8	7.0	10.7
	EI	2	11	5865.0	5310.3	6967.0	8.3	9.2
	BAv2	2	11	21042.8	20530.0	23386.6	4.4	5.5
	SAS	2	11	79265.8	77386.1	82131.1	2.0	3.1
	Svia	2	11	645487.0	625194.2	663217.5	2.0	3.3
	Te	2	11	25529.7	23523.1	27354.9	3.7	2.4
	HS	2	11	32564.0	30437.8	35507.4	3.7	1.8
	SVM	3	11	1531.6	1215.1	1801.2	10.5	6.2
	eM	3	11	3919.7	3280.7	4961.2	13.0	17.3
	EI	3	11	11889.8	11103.9	12535.8	4.0	5.9
	BAv2	3	11	27549.7	25524.5	28170.9	3.8	5.4
	SAS	3	11	158457.3	154079.8	168028.7	2.4	2.8
	Svia	3	11	1235627.5	1200462.2	1271183.7	1.7	2.2
	Te	3	11	39176.4	36239.9	42241.2	4.2	4.3
	HS	3	11	48995.8	47187.2	53333.3	4.2	6.7
	SVM	4	11	1534.3	1314.5	2035.1	12.8	12.8
	eM	4	11	7331.0	6256.0	8372.7	8.5	6.8
	EI	4	11	18891.3	17237.2	20539.5	5.1	4.0
	BAv2	4	11	26500.9	25456.5	27603.7	2.7	3.9
	SAS	4	11	293335.8	282210.8	298833.3	1.9	2.2
	Svia	4	11	2059771.2	2013698.0	2132528.5	1.5	0.9
	Te	4	11	48946.5	47830.4	53146.3	3.0	2.1
	HS	4	11	61192.8	58436.1	65810.1	3.4	3.7
Structural baseline	SVM	2	11	11998.4	10482.3	16338.5	17.5	30.5
	eM	2	11	21542.6	18939.8	34752.8	19.6	8.2
	EI	2	11	40288.1	37170.7	104809.1	42.9	5.6
	BAv2	2	11	664279.6	649083.1	1526834.6	35.1	1.3
	SAS	2	11	3303437.8	3227411.7	10427344.6	54.3	2.0
	Svia	2	11	809451.3	792811.5	1375504.1	20.0	2.1
	Te	2	11	579951.0	572158.0	1564025.0	44.2	2.1
	HS	2	11	684224.6	662907.3	1490940.3	32.4	2.6
	SVM	3	11	12223.6	10878.6	17716.3	15.4	11.9
	eM	3	11	20185.1	18281.8	40125.7	28.0	6.0
	EI	3	11	41895.6	38694.3	221899.3	94.0	2.8
	BAv2	3	11	662942.2	656173.9	1660129.6	39.9	0.7
	SAS	3	11	3402384.6	3291666.2	17235140.4	89.6	2.3
	Svia	3	11	861791.9	848557.3	2701666.7	54.0	1.1
	Te	3	11	599331.3	577722.5	2283726.2	67.6	1.7
	HS	3	11	695590.7	684050.9	1878476.1	44.3	3.1
	SVM	4	11	11171.2	10295.9	18134.5	21.5	20.1
	eM	4	11	21421.5	18927.0	51107.6	37.8	8.3
	EI	4	11	42370.2	41282.6	289709.4	113.6	6.5
	BAv2	4	11	667225.2	651861.8	1665426.2	39.8	1.7
	SAS	4	11	3542691.5	3435684.3	25610833.2	120.8	2.0
	Svia	4	11	923838.3	898205.5	5828258.0	108.3	1.9
	Te	4	11	614102.9	599360.5	2965674.7	86.0	1.8
	HS	4	11	709146.3	695223.8	2424193.7	59.6	2.3
Stochastic baseline	SVM	—	11	648.8	552.1	843.1	13.0	17.5
	eM	—	11	1310.0	1209.4	1667.4	9.8	13.0
	EI	—	11	2287.2	2038.2	3284.2	15.7	12.4
	BAv2	—	11	12945.3	11847.3	13665.6	3.9	4.3
	SAS	—	11	98011.1	95980.8	101164.7	2.0	3.6
	Svia	—	11	730312.1	709301.5	744261.7	1.4	1.5
	Te	—	11	32727.1	31569.5	34500.3	2.8	3.1
	HS	—	11	383391.1	370406.4	396289.2	2.2	2.4

Table 7
S7: JVM warm-up evidence per (SPL, approach, L) cell. The Mean/Median and Max/Median ratios across the eleven repetitions isolate cells in which a single high outlier inflates the mean while leaving the median and IQR almost unchanged. Cells with high CV% but low IQR% correspond to the warm-up fingerprint discussed in Section **??** and in the construct-validity threat in Section **??**.

Approach	SPL	L	Median (ms)	Max (ms)	CV (%)	IQR %	Mean / Median	Max / Median
Model Once, Generate Any	SVM	2	1389.6	1913.3	15.8	15.4	0.997	1.377
	eM	2	2773.9	3092.8	7.1	10.7	1.009	1.115
	EI	2	5865.0	6967.0	8.3	9.2	1.017	1.188
	BAv2	2	21042.8	23386.6	4.4	5.5	1.022	1.111
	SAS	2	79265.8	82131.1	2.0	3.1	1.007	1.036
	Svia	2	645487.0	663217.5	2.0	3.3	0.994	1.027
	Te	2	25529.7	27354.9	3.7	2.4	1.004	1.071
	HS	2	32564.0	35507.4	3.7	1.8	1.004	1.090
	SVM	3	1531.6	1801.2	10.5	6.2	1.000	1.176
	eM	3	3919.7	4961.2	13.0	17.3	0.998	1.266
	EI	3	11889.8	12535.8	4.0	6.0	0.995	1.054
	BAv2	3	27549.7	28170.9	3.8	5.4	0.984	1.023
	SAS	3	158457.3	168028.7	2.5	2.8	1.006	1.060
	Svia	3	1235627.5	1271183.7	1.7	2.2	0.998	1.029
	Te	3	39176.4	42241.2	4.2	4.3	0.994	1.078
	HS	3	48995.8	53333.3	4.2	6.7	1.017	1.089
	SVM	4	1534.3	2035.1	12.8	12.8	1.026	1.326
	eM	4	7331.0	8372.7	8.5	6.8	0.992	1.142
	EI	4	18891.3	20539.5	5.1	4.0	0.997	1.087
	BAv2	4	26500.9	27603.7	2.7	3.9	1.005	1.042
	SAS	4	293335.8	298833.3	1.9	2.2	0.993	1.019
	Svia	4	2059771.2	2132528.5	1.5	0.9	1.000	1.035
	Te	4	48946.5	53146.3	3.0	2.1	1.004	1.086
	HS	4	61192.8	65810.1	3.4	3.7	1.000	1.075
Structural baseline	SVM	2	11998.4	16338.5	17.5	30.5	1.071	1.362
	eM	2	21542.6	34752.8	19.6	8.2	1.025	1.613
	EI	2	40288.1	104809.1	42.9	5.6	1.136	2.601
	BAv2	2	664279.6	1526834.6	35.1	1.3	1.116	2.298
	SAS	2	3303437.8	10427344.6	54.3	2.1	1.198	3.157
	Svia	2	809451.3	1375504.1	20.0	2.1	1.061	1.699
	Te	2	579951.0	1564025.0	44.2	2.1	1.157	2.697
	HS	2	684224.6	1490940.3	32.4	2.6	1.103	2.179
	SVM	3	12223.6	17716.3	15.4	11.9	1.036	1.449
	eM	3	20185.1	40125.7	28.0	6.0	1.085	1.988
	EI	3	41895.6	221899.3	94.0	2.8	1.381	5.296
	BAv2	3	662942.2	1660129.6	39.9	0.7	1.137	2.504
	SAS	3	3402384.6	17235140.4	89.6	2.4	1.369	5.066
	Svia	3	861791.9	2701666.7	54.0	1.1	1.193	3.135
	Te	3	599331.3	2283726.2	67.6	1.8	1.254	3.810
	HS	3	695590.7	1878476.1	44.3	3.1	1.157	2.701
	SVM	4	11171.2	18134.5	21.5	20.1	1.134	1.623
	eM	4	21421.5	51107.6	37.8	8.3	1.122	2.386
	EI	4	42370.2	289709.4	113.6	6.5	1.546	6.838
	BAv2	4	667225.2	1665426.2	39.8	1.7	1.136	2.496
	SAS	4	3542691.5	25610833.2	120.8	2.0	1.558	7.229
	Svia	4	923838.3	5828258.0	108.3	1.9	1.479	6.309
	Te	4	614102.9	2965674.7	86.0	1.8	1.344	4.829
	HS	4	709146.3	2424193.7	59.6	2.3	1.223	3.418
Stochastic baseline	SVM	—	648.8	843.1	13.0	17.5	1.007	1.300
	eM	—	1310.0	1667.4	9.8	13.0	1.042	1.273
	EI	—	2287.2	3284.2	15.7	12.4	1.059	1.436
	BAv2	—	12945.3	13665.6	3.9	4.3	0.998	1.056
	SAS	—	98011.1	101164.7	2.0	3.6	1.006	1.032
	Svia	—	730312.1	744261.7	1.4	1.5	1.001	1.019
	Te	—	32727.1	34500.3	2.8	3.1	1.006	1.054
	HS	—	383391.1	396289.2	2.2	2.4	0.999	1.034

Table 8
S8: Pairwise Wilcoxon signed-rank results, Vargha–Delaney A_{12} effect sizes, and Benjamini–Hochberg corrected p-values for every metric and (SPL, L) combination. Approach pairs MO vs Struct and MO vs Stoch back the cross-approach claims in Section **??**.

Metric	SPL	Approach A	Approach B	L	N	W	p_{BH}	A_{12}	Winner
Edge coverage (%)	SVM	MO	Struct	$L = 2$	11	0	n.s.	0.50	tie
	SVM	MO	Stoch	$L = 2$ vs —	11	0	n.s.	0.50	tie
	SVM	Struct	Stoch	$L = 2$ vs —	11	0	n.s.	0.50	tie
	SVM	MO	Struct	$L = 3$	11	0	n.s.	0.50	tie
	SVM	MO	Struct	$L = 4$	11	0	n.s.	0.50	tie
	eM	MO	Struct	$L = 2$	11	0	0.001	1.00	MO
	eM	MO	Stoch	$L = 2$ vs —	11	0	n.s.	0.50	tie
	eM	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Stoch
	eM	MO	Struct	$L = 3$	11	0	0.001	1.00	MO
	eM	MO	Struct	$L = 4$	11	0	0.001	1.00	MO
	EI	MO	Struct	$L = 2$	11	0	0.001	1.00	MO
	EI	MO	Stoch	$L = 2$ vs —	11	0	n.s.	0.50	tie
	EI	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Stoch
	EI	MO	Struct	$L = 3$	11	0	n.s.	0.50	tie
	EI	MO	Struct	$L = 4$	11	0	n.s.	0.50	tie
	BAv2	MO	Struct	$L = 2$	11	0	0.001	1.00	MO
	BAv2	MO	Stoch	$L = 2$ vs —	11	0	n.s.	0.50	tie
	BAv2	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Stoch
	BAv2	MO	Struct	$L = 3$	11	0	0.001	1.00	MO
	BAv2	MO	Struct	$L = 4$	11	0	0.001	0.00	Struct
	SAS	MO	Struct	$L = 2$	11	0	0.001	1.00	MO
	SAS	MO	Stoch	$L = 2$ vs —	11	0	n.s.	0.50	tie
	SAS	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Stoch
	SAS	MO	Struct	$L = 3$	11	0	0.001	1.00	MO
	SAS	MO	Struct	$L = 4$	11	0	0.001	1.00	MO
	Svia	MO	Struct	$L = 2$	11	0	0.001	1.00	MO
	Svia	MO	Stoch	$L = 2$ vs —	11	0	0.001	1.00	MO
	Svia	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Stoch
	Svia	MO	Struct	$L = 3$	11	0	0.001	1.00	MO
	Svia	MO	Struct	$L = 4$	11	0	0.001	1.00	MO
	Te	MO	Struct	$L = 2$	11	0	0.001	1.00	MO
	Te	MO	Stoch	$L = 2$ vs —	11	0	0.001	1.00	MO
	Te	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Stoch
	Te	MO	Struct	$L = 3$	11	0	0.001	1.00	MO
	Te	MO	Struct	$L = 4$	11	0	0.001	1.00	MO
	HS	MO	Struct	$L = 2$	11	0	0.001	1.00	MO
	HS	MO	Stoch	$L = 2$ vs —	11	0	0.001	1.00	MO
	HS	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Stoch
	HS	MO	Struct	$L = 3$	11	0	0.001	1.00	MO
	HS	MO	Struct	$L = 4$	11	0	0.001	1.00	MO
Peak heap memory (MB)	SVM	MO	Struct	$L = 2$	11	0	0.001	1.00	Struct
	SVM	MO	Stoch	$L = 2$ vs —	11	0	0.001	1.00	Stoch
	SVM	Struct	Stoch	$L = 2$ vs —	11	0	0.001	1.00	Stoch
	SVM	MO	Struct	$L = 3$	11	0	0.001	1.00	Struct
	SVM	MO	Struct	$L = 4$	11	0	0.001	1.00	Struct
	eM	MO	Struct	$L = 2$	11	0	0.001	1.00	Struct
	eM	MO	Stoch	$L = 2$ vs —	11	0	0.001	1.00	Stoch
	eM	Struct	Stoch	$L = 2$ vs —	11	0	0.001	1.00	Stoch
	eM	MO	Struct	$L = 3$	11	0	0.001	1.00	Struct
	eM	MO	Struct	$L = 4$	11	0	0.001	1.00	Struct
	EI	MO	Struct	$L = 2$	11	0	0.001	1.00	Struct
	EI	MO	Stoch	$L = 2$ vs —	11	0	0.001	1.00	Stoch
	EI	Struct	Stoch	$L = 2$ vs —	11	0	0.001	1.00	Stoch
	EI	MO	Struct	$L = 3$	11	0	0.001	1.00	Struct
	EI	MO	Struct	$L = 4$	11	0	0.001	1.00	Struct
	BAv2	MO	Struct	$L = 2$	11	0	0.001	1.00	Struct
	BAv2	MO	Stoch	$L = 2$ vs —	11	0	0.001	0.00	MO
	BAv2	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Struct
	BAv2	MO	Struct	$L = 3$	11	0	0.001	1.00	Struct
	BAv2	MO	Struct	$L = 4$	11	0	0.002	0.97	Struct
	SAS	MO	Struct	$L = 2$	11	0	n.s.	0.41	MO
	SAS	MO	Stoch	$L = 2$ vs —	11	0	0.001	0.00	MO
	SAS	Struct	Stoch	$L = 2$ vs —	11	0	0.001	0.00	Struct
	SAS	MO	Struct	$L = 3$	11	0	0.001	1.00	Struct
	SAS	MO	Struct	$L = 4$	11	0	0.001	1.00	Struct
	Svia	MO	Struct	$L = 2$	11	0	0.001	1.00	Struct

Table 9
S9: Per-SPL edge-count distribution. The range ratio $\max / \min$ quantifies the within-SPL variation in product complexity. Section **??** relates this range to the per-SPL correlation: SPLs with a narrower edge range (e.g., Bank Account, $3.3\times$) typically produce weak per-SPL correlations, while SPLs with a wider range produce sharper trends.

SPL	Approach	N	Min	Median	Max	Range ratio
SVM	MO	12	5	8	14	2.80
SVM	Struct	12	3	6	11	3.67
SVM	Stoch	12	5	8	14	2.80
eM	MO	23	19	25	33	1.74
eM	Struct	23	12	16	21	1.75
eM	Stoch	23	19	25	33	1.74
El	MO	42	28	38	51	1.82
El	Struct	42	20	28	38	1.90
El	Stoch	42	28	38	51	1.82
BAv2	MO	498	16	33	53	3.31
BAv2	Struct	498	10	20	33	3.30
BAv2	Stoch	498	16	33	53	3.31
SAS	MO	2664	22	56	85	3.86
SAS	Struct	2664	14	41	63	4.50
SAS	Stoch	2664	22	56	85	3.86
Svia	MO	400	447	543	605	1.35
Svia	Struct	400	409	505.5	567	1.39
Svia	Stoch	400	447	543	605	1.35
Te	MO	400	28	38.5	65	2.32
Te	Struct	400	26	36.5	63	2.42
Te	Stoch	400	28	38.5	65	2.32
HS	MO	416	60	70	75	1.25
HS	Struct	416	58	68	73	1.26
HS	Stoch	416	60	70	75	1.25